Proof

Let the random variable X assumes the values $x_1, x_2 \ldots x_n$ with corresponding probabilities $p_1, p_2 \ldots . p_n$, and the random variable y assume the values $y_1, y_2 \ldots \ldots . y_m$ with corresponding probabilities $p_1, p_2 \ldots . p_m$

By definition,

$$E(X) = \sum x_i p_i \qquad \text{where } \sum p_i = 1$$

$$E(Y) = \sum y\, jpj \qquad \sum p_j = 1$$

Now,
$$E(X+Y) = \sum_{i=1}^{n} \sum_{j=1}^{m} p_{ij}(x_i + y_j)$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{m} p_{ij} x_i + \sum_{j=1}^{n} p_{ij}(x_i + y_j)$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{m} p_{ij} x_i + \sum_{i=1}^{n} \sum_{i=1}^{m} p_{ij} y_j$$

$$= \sum_{i=1}^{n} x_i p_i + \sum_{j=1}^{m} y_j p_j$$

$$= E(X) + E(Y)$$

2. Statement for Continuous random variable

Let X and Y are two continuous random variables with probability density functions $f(x)$ and $f(y)$ respectively. Then $E(X + Y) = E(X) + E(Y)$

Proof:

We know that

$$E(X) = \int_{-\infty}^{\infty} x f(x)\, dx \qquad \text{and}$$

$$E(Y) = \int_{-\infty}^{\infty} y f(y)\, dy$$

$$E(X+Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x+y) f(x, y)\, dx dy$$

$$= \int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} f(x, y)\, dy \right) dx + \int_{-\infty}^{\infty} y \left(\int_{-\infty}^{\infty} f(x, y)\, dx \right) dy$$

$$= \int_{-\infty}^{\infty} x f(x)\, dx + \int_{-\infty}^{\infty} y f(y)\, dy$$

$$= E(X) + E(Y)$$